Ali Mortazavi

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Summary

PhD candidate in online learning and game theory, with publications at top conferences (**NeurIPS**, **AISTATS**, **ACM EC**). Seeking Applied Scientist and research roles in ML and optimization.

Education

Ph.D. in Computer Science, University of Victoria — Jan 2021 – Dec 2026 (Expected)

M.Sc. in Artificial Intelligence, Amirkabir University of Technology (Tehran Polytechnic) — Sept 2017 – Sept 2020

B.Sc. in Computer Software Engineering, Amirkabir University of Technology — Sept 2013 – Sept 2017

Research Experience

PhD **Research Assistant**, Machine Learning Theory Group, University of Victoria — Jan 2021 – Present

- **Reputation-Based Sequential Forecasting** (ACM EC 2025) `(Game Theory, Information Elicitation)`
  - Co-designed the first truthful mechanism that combines predictions from many strategic forecasters over time, where the average error of the combined prediction is at most the best forecaster's plus $O(1/\sqrt{T})$ over T rounds. This result settled an open question of whether achieving this guarantee is possible at all.
  - Extended the mechanism to the partial information setting (multi-armed bandit), the first such result for this setting.
- **Aggregating Advice from Selection-Seeking Forecasters** (AISTATS 2024) `(Incentives, Partial feedback)`
  - Used martingale concentration arguments to prove that the worse performance guarantee (higher regret) of the only known (at the time) strategy-proof forecaster selector algorithm is an inherent limitation and not an artifact of loose analysis.
- **Adaptive Decision-Making While Preserving Group Fairness** (NeurIPS 2021) `(FTRL, Fairness)`
  - Showed that a broad family of anytime algorithms (e.g., Decreasing Hedge) provably preserves equal error rates across groups at every round instead of only the last, without knowing group sizes.

Technical Skills

- **Mathematical Methods:** Linear and convex optimization (LP formulation, duality, KKT), Online learning, Multi-armed bandits and adaptive experimentation, Probability and concentration inequalities, Game theory, mechanism and auction design
- **Programming:** Python, Java, C++, NumPy, Pandas, Matplotlib, TensorFlow, scikit-learn, PyTorch, Git, AI coding agents

Publications

- Junpei Komiyama, Nishant Mehta, and Ali Mortazavi. "No-Regret Incentive-Compatible Online Learning under Exact Truthfulness with Non-Myopic Experts" *ACM Conference on Economics and Computation* (**EC**), 2025.
- Ali Mortazavi, Junhao Lin, and Nishant Mehta. "On the Price of Exact Truthfulness in Incentive-Compatible Online Learning with Bandit Feedback" *International Conference on Artificial Intelligence and Statistics* (**AISTATS**), 2024.
- Cristóbal Guzmán, Nishant Mehta, and Ali Mortazavi. "Best-Case Lower Bounds in Online Learning" *Conference on Neural Information Processing Systems* (**NeurIPS**), 2021.

Projects

- **Calorie Tracker** (Try it) `(Python, JavaScript, Supabase, Postgres)`

  Built and deployed a web app with AI coding agents to track food and macros, with a barcode scanner and simple food descriptions that make it quick to use.

Teaching

Teaching Assistant, Computer Science Department, University of Victoria — Jan 2021 – Present

Courses include: Machine Learning Theory, Data Mining, Algorithms and Data Structures II, Collective Decision-Making.

Awards and Service

- Best Reviewer Award (top 4%), International Conference on Artificial Intelligence and Statistics (AISTATS) — 2025
- Reviewer for ICML 2025, AISTATS 2025, NeurIPS 2024, and student volunteer at STOC 2024
- University of Victoria PhD Fellowship, Graduate TA Award, and competitive graduate scholarships — 2021–2024
- Organizer of a bi-weekly tea event called "**More Tea Community**" — Feb 2026 – Present